

RV College of Engineering
DAA Experiential Learning

Synopsis

2025-26



Title of the Project

**Stability Aware Decision Algorithm for
Multi Agent System**

Team No: 7

Team Details

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1. Introduction

The Stability-Aware Decision Algorithm (SADA) is a novel approach designed to improve decision-making in multi-agent systems operating in dynamic environments. In conventional multi-agent pathfinding and decision systems, agents optimize for shortest paths and collision avoidance. However, these approaches often lead to unstable behavior, where agents frequently change actions due to local disturbances, resulting in oscillations commonly referred to as *decision jitter*.

This instability not only reduces efficiency but also increases computational overhead and coordination difficulty, especially in dense environments. SADA addresses this limitation by introducing a stability-aware cost component into the decision-making process.

The algorithm incorporates a history-based penalty mechanism that discourages frequent action changes. By tracking recent actions and penalizing excessive fluctuations, SADA ensures smoother and more consistent trajectories while maintaining near-optimal path efficiency.

This project focuses on designing, implementing, and evaluating SADA within a simulated multi-agent grid environment, demonstrating how algorithmic modifications can significantly improve system stability without requiring complex global coordination.

2. Literature Survey

1. Incremental CBS: Adaptive Multi-Agent Pathfinding in Dynamic Environments

Abstract- This study addresses the limitations of traditional Conflict-Based Search (CBS) in dynamic environments by introducing Incremental Conflict-Based Search (ICBS). ICBS improves efficiency by updating only affected agent paths using a dependency graph and dynamic monitoring. Experimental results show it outperforms CBS, making it suitable for real-time applications like robotics and warehouse systems.

Xing, X., Wang, B., & Cheng, J. (2024). Incremental CBS: Adaptive Multi-Agent Pathfinding in Dynamic Environments. *2024 7th International Conference on*

Robotics, Control and Automation Engineering (RCAE), 164-169.
<https://doi.org/10.1109/rcae62637.2024.10834259>.

2. A hybrid optimization algorithm for multi-agent dynamic planning with guaranteed convergence in probability

Abstract- This paper proposes an improved whale optimization algorithm for multi-agent path planning in complex environments, addressing issues like local optima and premature convergence. The method integrates chaotic initialization, reverse search, and differential evolution to enhance global convergence. Simulation results with multi-UAVs show superior performance, robustness, and smoother path generation compared to existing methods.

Zhang, Y., Zhu, Y., Li, H., & Wang, J. (2024). A hybrid optimization algorithm for multi-agent dynamic planning with guaranteed convergence in probability. *Neurocomputing*, 592, 127764.
<https://doi.org/10.1016/j.neucom.2024.127764>.

3. Decentralized Multi-Agent Motion Planning in Dynamic Environments

Abstract- This paper presents a decentralized multi-agent motion planning algorithm for dynamic environments using a learning-based framework. Each agent plans paths considering kinodynamic constraints and uses a game-theoretic approach to avoid conflicts with others. The method updates paths online to prevent collisions and freezing issues, showing effective performance in simulations.

Netter, J., & Vamvoudakis, K. (2023). Decentralized Multi-Agent Motion Planning in Dynamic Environments. *2023 American Control Conference (ACC)*, 1655-1660.
<https://doi.org/10.23919/acc55779.2023.10156024>.

4. Multi-Agent Path Finding Among Dynamic Uncontrollable Agents with Statistical Safety Guarantees

Abstract- This paper proposes a modified Enhanced Conflict-Based Search (ECBS) algorithm to handle uncertain behavior of uncontrollable agents in dynamic environments. It combines learned motion prediction with conformal prediction to

quantify uncertainty and ensure safer path planning. The approach provides statistical guarantees for collision avoidance and shows improved performance in both one-shot and lifelong MAPF scenarios.

Strawn, K., Phan, T., Wang, E., Ayanian, N., Koenig, S., & Lindemann, L. (2025). Multi-Agent Path Finding Among Dynamic Uncontrollable Agents with Statistical Safety Guarantees. *ArXiv*, *abs/2507.22282*.
<https://doi.org/10.48550/arxiv.2507.22282>.

5. Multi-Agent Distributed Optimal Control for Tracking Large-Scale Multi-Target Systems in Dynamic Environments

Abstract- This paper addresses multi-agent motion coordination for tracking multiple targets in dynamic environments with obstacles. It uses density-based planning and adaptive optimal control to guide agents while ensuring collision avoidance. A distributed Voronoi-based algorithm enables effective coordination, with simulations demonstrating its performance.

Abdulghafoor, A., & Bakolas, E. (2023). Multi-Agent Distributed Optimal Control for Tracking Large-Scale Multi-Target Systems in Dynamic Environments. *IEEE Transactions on Cybernetics*, *54*, 2866-2879.
<https://doi.org/10.1109/tcyb.2023.3302288>.

3. Problem Definition

Multi-agent systems often operate in environments where each agent must make decisions based on local information while avoiding collisions and reaching individual goals. Traditional approaches rely on cost functions that prioritize distance minimization and collision avoidance. However, these methods fail to account for temporal consistency in decisions.

As a result, agents frequently switch between actions when faced with small environmental changes, leading to oscillatory behavior and inefficient movement patterns.

The problem can be formally defined as:

Given multiple agents operating in a shared environment, how can each agent select actions that are not only optimal in the current step but also stable across time, minimizing unnecessary decision changes?

The challenge lies in integrating stability into decision-making without significantly increasing computational complexity or degrading path optimality.

4. Objectives

Primary Objectives

- To design a stability-aware decision algorithm for multi-agent systems.
- To reduce decision jitter by introducing a flip-based penalty mechanism.
- To maintain high efficiency in pathfinding while improving trajectory smoothness.
- To implement the algorithm in a grid-based simulation environment.

Secondary Objectives

- To analyze the trade-off between stability and optimality.
- To perform comparative evaluation with baseline algorithms.
- To visualize agent behavior through simulations and animations.
- To ensure scalability for increasing numbers of agents.

5. Methodology

Approach:

The project follows a modular and phased approach, where each component is developed, tested, and integrated systematically.

The core idea is to modify the traditional cost function by adding a stability term:

- Each agent maintains a history of past actions.
- A flip count is computed based on how often actions change.
- A penalty proportional to this flip count is added to the decision cost.

This ensures agents prefer consistent actions over erratic switching.

Procedures:

1. Environment Setup
 - Create grid world with obstacles and goals.
 - Initialize multiple agents.
2. Baseline Implementation
 - Implement standard cost function (distance + collision).
3. History Tracking
 - Store last K actions using a deque.
 - Compute flip count based on action changes.
4. SADA Integration
 - Modify cost function to include stability penalty.
 - Tune parameters (β , K).
5. Simulation Execution
 - Run multiple scenarios with varying agent counts and environments.
6. Evaluation
 - Measure flip rate, success rate, path length, and smoothness.
7. Visualization
 - Generate side-by-side comparison (baseline vs SADA).

This structured approach ensures proper validation and realistic simulation of multi-agent behavior.

6. Expected Outcome

The project is expected to deliver:

- A fully functional multi-agent simulation system.
- Significant reduction in decision jitter.
- Improved trajectory smoothness and stability.
- Comparable goal success rate ($>90\%$).
- Controlled increase in path length ($\leq 15\%$ overhead).
- Visual demonstration showing clear difference between:
 - Baseline (unstable)
 - SADA (stable)

Additionally, the system will provide quantitative metrics such as:

- Flip rate reduction
- Velocity variance decrease
- Scalability across multiple agents

7. Conclusion

The Stability-Aware Decision Algorithm (SADA) introduces a simple yet powerful modification to traditional multi-agent decision systems by incorporating temporal stability into the cost function.

Unlike conventional approaches that focus solely on optimality, SADA balances efficiency with behavioral consistency, leading to smoother and more reliable agent movements. The algorithm remains computationally lightweight, decentralized, and adaptable to various environments.

This project demonstrates that even minor enhancements in cost formulation can significantly improve real-world system performance. It also opens avenues for

future work in areas such as reinforcement learning integration, adaptive parameter tuning, and real-world robotic navigation systems.